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ORIGINAL ARTICLE

Design and multi-objective optimization of amorphous alloy PMSM for EVs over a full driving cycle

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Abstract
Electric vehicles (EVs) face significant challenges when it comes to extending their mileage and reducing manufacturing costs. Enhancing the performance of the driving motor is an indispensable approach to address these issues. A new amorphous alloy permanent-magnet synchronous machine (PMSM) and a driving-cycle oriented NSGA-II (DCO-NSGA-II) optimization design method are proposed in this paper. Based on the China light-duty vehicle test cycle for passenger cars (CLTC-P), the full-speed range characteristics of the amorphous alloy PMSM were analyzed. Representative operating points and corresponding weighting factors were extracted via clustering analysis, and key design parameters were identified using the sensitivity analysis method. A multi-objective optimization model was established targeting the output torque, torque ripple, cycle efficiency, and effective material cost. The Pareto optimal set was obtained using the nondominated sorting genetic algorithm II (NSGA-II). Finite element analysis (FEA) showed that the optimized motor achieved a 3.44% increase in peak torque, a 1.3% reduction in torque ripple, a 1.5% improvement in overall driving cycle efficiency, and a 1.2% reduction in effective cost compared with the initial design. Finally, a prototype was manufactured and tested, confirming the effectiveness of the proposed approach in realizing coordinated optimization of performance and economics.

Keywords Amorphous alloy · Permanent magnet machine · Multi-objective optimization · Driving cycle

1 Introduction

In recent years, the global electric vehicle (EV) industry has experienced rapid growth, with market penetration rates climbing steadily. As a core propulsion component, the efficiency of the driving motor directly affects the driving range and economic feasibility of electric vehicles. Consequently, the industry has an urgent demand for low-loss, high-efficiency motors [1]. Breakthroughs in performance largely depend on innovations in new materials, advanced manufacturing processes, and novel motor topologies [2].

Amorphous alloy, as a new type of soft magnetic material with excellent magnetic properties including high magnetic permeability, low specific loss, and low coercivity, has attracted widespread attention in the field of electrical machines [3]. Tong et al. demonstrated that substituting silicon steel with amorphous alloy in the stator core of a permanent-magnet synchronous motor (PMSM) can significantly reduce iron loss and improve efficiency in the high-speed region, although its advantages may diminish under low-speed and high-torque condition [4]. Furthermore, Zhang et al. addressed the issue of rising copper loss in an amorphous motor for low-frequency application by optimizing the stator and rotor structures, effectively improving comprehensive efficiency [5]. These studies indicate that the performance advantages of amorphous alloys can only be fully exploited through careful structural optimization.

The optimal design of motors generally involves constructing surrogate models based on the nonlinear correlation between structural parameters and performance, and then adopting optimization algorithms to obtain the optimal parameter combination [6]. For instance, Wang et al. adopted a response surface methodology to establish models of torque density and rotor stress, and conducted multi-objective optimization with improved algorithms [7].

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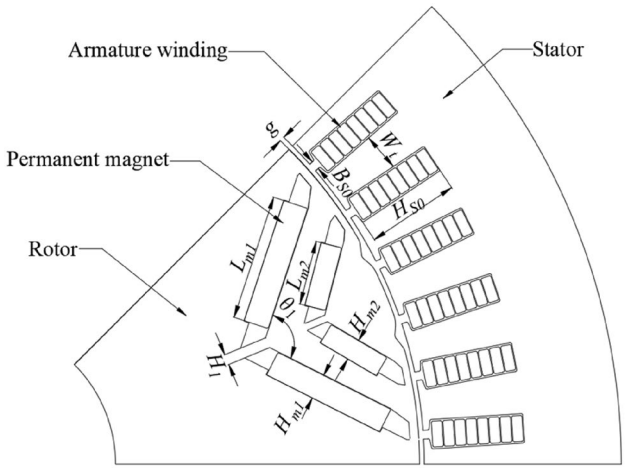


Fig. 1 Motor topology and definition of parameters

Table 1 Main technical specifications

Parameter	Value	Parameter	Value
Rated power/kW	82	Peak power/kW	220
Rated speed/rpm	6000	Peak speed/rpm	18,000
Rated torque/(N·m)	160	Peak torque/(N·m)	360
Rated voltage/V	630	Axial length/mm	100
Stator OD/mm	220	Rotor ID/mm	50
Stator slots	48	Rotor poles	8

However, most existing research focuses on optimization under rated conditions. Although it is computationally efficient, this approach may not deliver comprehensive optimal performance for driving motors, which need to accommodate complex and variable driving demands in both urban and highway scenarios. Optimization based on a full driving cycle is more likely to achieve superior adaptability [8]. Sun et al. constructed a hybrid operating condition by integrating multiple standard cycles, and realized the full-operating-condition optimization design of a motor via the improved particle swarm optimization algorithm [9]. However, their optimization objective is limited to torque characteristics, neglecting the influence of other performance factors.

To improve motor performance, a new amorphous alloy PMSM and a corresponding driving-cycle oriented NSGA-II (DCO-NSGA-II) strategy based on the full vehicle driving cycle is proposed in this paper. Unlike conventional driving-cycle-based optimization methods that only extract representative operating points, the proposed approach further incorporates weighting factors derived from driving cycle into the objective evaluation, enabling the optimization to reflect the actual distribution characteristics of the operating conditions. Furthermore, Sobol sensitivity analysis is integrated to reduce the dimensionality of design variables, improving computational efficiency and robustness.

The remainder of this paper is organized as follows. First, the topology of the amorphous alloy PMSM is proposed and key design parameters are screened via Sobol sensitivity analysis in Sect. 2. Second, representative operating points are extracted from the China light-duty vehicle test cycle for passenger cars (CLTC-P) cycle using the K-means clustering method in Sect. 3. The NSGA-II algorithm is used to solve for the optimal solution in Sect. 4. The validity of the proposed method is verified by simulation and experimental prototype experiments in Sect. 5 and Sect. 6, respectively. Finaly, some conclusions are given in Sect. 7.

2 Proposed amorphous alloy PMSM

2.1 Motor topology and specifications

The proposed amorphous alloy PMSM topology and its optimization variables are illustrated in Fig. 1. A double-V interior permanent magnet rotor structure was adopted to satisfy the requirements for high torque density and wide flux-weakening capability. The stator core was fabricated from amorphous alloy laminations. Considering high-speed operation and manufacturing feasibility, an 8-pole/48-slot configuration with eight-layer hairpin windings was adopted. The grooves on the rotor surface are auxiliary structures designed to improve the air-gap flux density waveform and suppress cogging torque as well as torque ripple. These structures were optimized during the initial design stage using finite element analysis and remained fixed in the subsequent optimization process to avoid excessive computational complexity.

To fully leverage the advantages of amorphous alloy and ensure optimization reliability, the stator outer diameter, axial length, and slot fill factor were kept constant. Ten parameters were initially selected as optimization variables.

The main technical specifications of the proposed amorphous alloy PMSM are listed in Table 1.

2.2 Amorphous alloy material characteristics

Amorphous alloy materials possess a non-crystalline atomic structure, which results in low magnetic anisotropy, low coercivity, and high electrical resistivity. The amorphous alloy used in this study has an electrical resistivity of 130 μΩ·cm, a saturation flux density B_S of 1.56 T, and a coercivity H_c of 3.7 A/m. The B-H curve and B-P curve are shown in Fig. 2a and b, respectively. According to the B–H and B–P characteristics, amorphous alloys can significantly reduce hysteresis loss and eddy-current losses under alternating magnetic fields.

Amorphous alloys were fabricated through rapid solidification, forming thin ribbons with a typical thickness of approximately 0.025 mm. According to classical

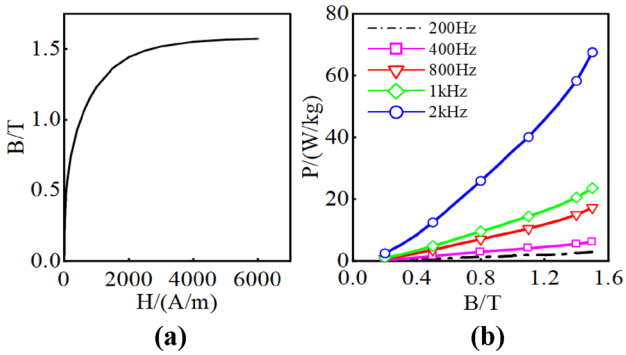


Fig. 2 Amorphous alloy characteristics: **a** B-H curve; **b** B-P curve

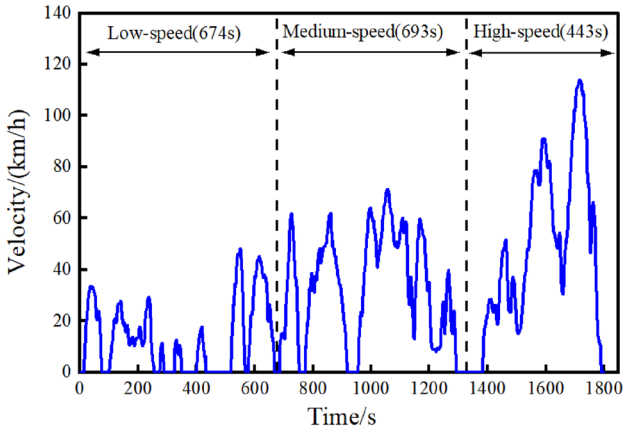


Fig. 3 CLTC-P vehicle driving cycle

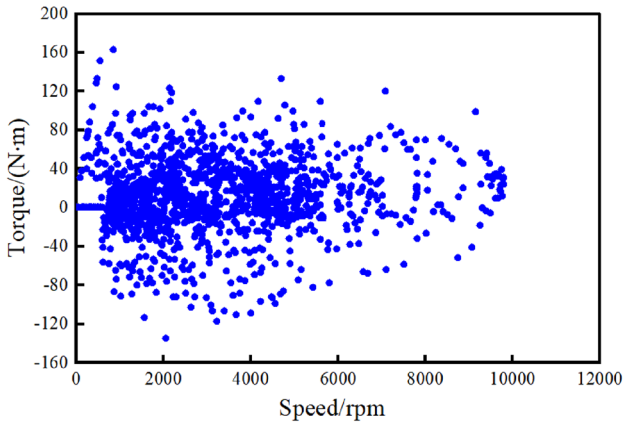


Fig. 4 Motor speed-torque distribution

eddy-current theory, eddy-current loss is proportional to the square of the lamination thickness. The reduced thickness, combined with higher electrical resistivity, significantly mitigates eddy-current losses, especially under high-frequency operating conditions [10].

However, the saturation flux density of amorphous alloys is lower than that of conventional silicon steel. Therefore, the motor topology must be meticulously optimized to fully exploit the advantages of this material.

3 Driving cycle analysis

Constructed from large-scale real-world road driving data nationwide, the CLTC-P reflects the unique traffic conditions and driving characteristics of China, and serves as the core basis for evaluating vehicle energy consumption and emissions [11]. The CLTC-P vehicle driving cycle is shown in Fig. 3. This cycle includes low-speed, medium-speed, and high-speed operating conditions, with a total duration of 1800 s, a maximum speed of 114.0 km/h, and an average speed of 28.9 km/h.

Under actual driving conditions, the operating state of the vehicle can be described by Eqs. (1) and (2).

$$T_e = \frac{r}{\eta_T i_0}\left(fmg\cos\theta + \frac{C_D A v^2}{21.25} + \delta ma + mg\sin\theta\right) \tag{1}$$

$$n = \frac{v i_0}{0.377 r} \tag{2}$$

Where T_e is the motor torque; r is the wheel radius; i_0 is the transmission ratio; η_T is the transmission efficiency; f is the rolling resistance coefficient; m is the vehicle mass; α is the road slope angle; C_D is the drag coefficient; A is the frontal area; δ is the rotational mass conversion coefficient; v is the vehicle speed, and n is the motor speed.

Based on the above equations, the speed-torque distribution of the motor under the CLTC-P cycle was obtained, as shown in Fig. 4. When the motor output torque is negative, the driving motor operates in the regenerative braking mode. When both the speed and torque are zero, the motor remains stationary. Neither condition consumes electric energy. Therefore, only the operating points of the amorphous alloy PMSM in the motoring mode are selected as the research range for energy efficiency optimization.

Due to the high dimensionality and randomness of real driving data, directly performing optimization over all of the operating points leads to excessively high computational costs. To address this issue, the K-means clustering algorithm was adopted to extract representative operating points from the CLTC-P cycle. To determine the optimal number of clusters, the silhouette coefficient (SC) and Davies–Bouldin index (DBI) were employed as evaluation metrics. The SC measures the compactness and separation of clusters, with higher values indicating better performance; and the DBI evaluates cluster dispersion, with lower values representing

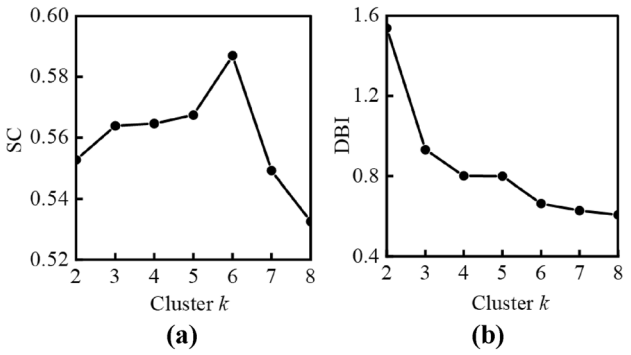


Fig. 5 Clustering evaluation indicators: **a** silhouette coefficient; **b** Davies-Bouldin index

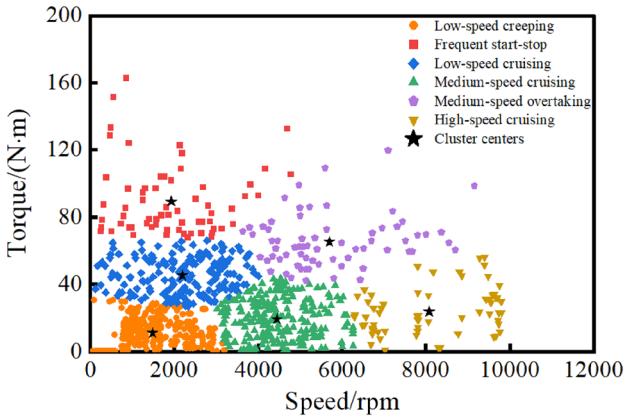


Fig. 6 Classification of motor operating conditions

Table 2 Representative operating points for CLTC-P

Operating Phase	Speed /(rpm)	Torque /(*N*·m)	Proportion /(%)
Low-speed creeping	1420	10.77	29.6
Frequent start-stop	1794	88.98	7.3
Low-speed cruising	2113	45.18	18.8
Medium-speed cruising	4391	19.18	29.1
Medium-speed overtaking	5615	64.84	7.7
High-speed cruising	8073	23.59	7.5

improved clustering quality [12]. The clustering evaluation results are shown in Fig. 5. These results show that six clusters achieve the best balance between clustering accuracy and computational efficiency, because further increasing the number of clusters yields only marginal improvements while greatly increasing the computational cost.

Accordingly, the CLTC-P cycle was categorized into six typical operating conditions, including low-speed creeping, frequent start-stop, low-speed cruising, medium-speed cruising, medium-speed overtaking, and high-speed cruising. The classification of the motor operating conditions is shown in Fig. 6.

The resulting cluster center operating points and weight proportions are given in Table 2.

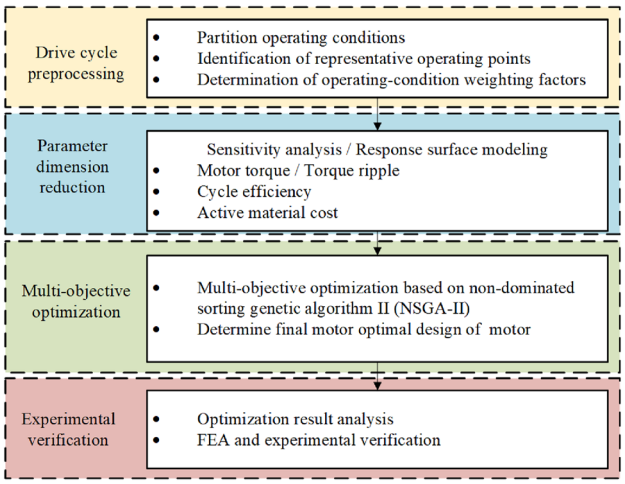


Fig. 7 Multi-objective optimization framework

4 Multi-objective optimization analysis

Based on the initial performance analysis, sensitive design parameters were identified. Accordingly, a systematic framework was constructed to guide the multi-objective optimization process via the proposed DCO-NSGA-II, as illustrated in Fig. 7.

4.1 Sensitivity analysis of optimization variables

To reduce the hardware requirements and time consumption of the optimization process, Sobol sensitivity analysis was used to determine the effect of each variable on the output torque, torque ripple, cycle efficiency, and active material cost (AMC). The sensitivity is defined as:

$$S_{Ti} = \frac{D_t - D_{-i}}{D_t} \tag{3}$$

Where S_{Ti} is the global sensitivity of variable x_i, D_t is the sum of the variances for the variable x_i, and D_{-i} is the total variance of the remaining terms. As a variance-based global method, Sobol sensitivity analysis quantifies both the main and interaction effects without relying on linearity assumptions. To mitigate the high computational cost associated with its large sampling requirements, an optimized Latin hypercube sampling (OLHS) strategy was adopted.

The sensitivity results are shown in Fig. 8. They indicate that the output torque is predominantly affected by the air gap length and magnet length, while the torque ripple is sensitive to the magnet angle and slot opening width. The cost is largely driven by the magnet dimensions, and the cycle efficiency is most sensitive to the air gap length and stator tooth width.

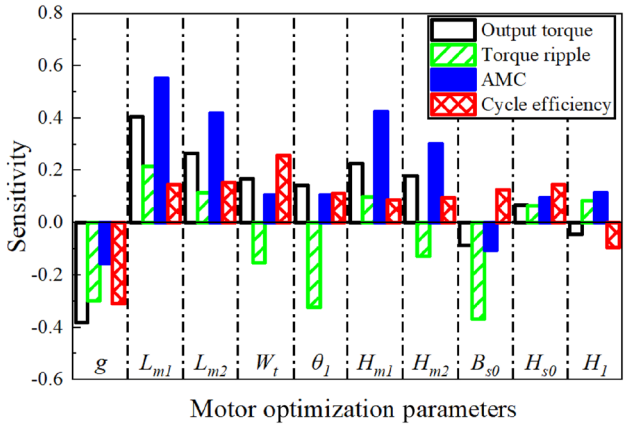


Fig. 8 Parameter sensitivity analysis

4.2 Optimization objectives

The average torque was defined as the mean peak torque at the rated speed. This metric was utilized to reflect the maximum output capacity of the motor, ensuring that the optimized design met performance requirements under high-load conditions.

Torque ripple was introduced as an objective to ensure stable operation and improved ride comfort.

$$T_{ripple} = \frac{T_{\max} - T_{\min}}{T_{avg}} \times 100\% \tag{4}$$

Where T_{max}, T_{min}, and T_{avg} are the maximum, minimum, and average torque, respectively.

The active material cost was calculated from the mass and the current market price as follows:

$$AMC = C_{AA}m_{AA} + C_{PM}m_{PM} + C_{Cu}m_{Cu} + C_{Fe}m_{Fe} \tag{5}$$

Where m_{AA}, m_{PM}, m_{Cu}, and m_{Fe} are the masses of the amorphous alloy, permanent magnets, copper, and silicon steel laminations, respectively. C_{AA}, C_{PM}, C_{Cu}, and C_{Fe} are the unit prices of the amorphous alloy, permanent magnets, copper, and silicon steel sheets, which are 4.00 USD/kg, 57.18 USD/kg, 8.86 USD/kg, and 2.86 USD/kg, respectively. The material prices used in the cost model are based on commercial quotations from material suppliers collected during the research period. The price benchmarks for permanent magnets, electrolytic copper, and electrical silicon steel were cross-verified with publicly available authoritative spot price data from the Shanghai Metals Market (SMM), which serves as a widely referenced industrial database for non-ferrous metals and magnetic materials. The price of the amorphous alloy was provided by the cooperative material manufacturer, which is consistent with the mainstream market price range of similar products. The relative price ratios of different materials remained stable over the research timeframe, which ensures the validity and comparability of the multi-objective optimization results.

Table 3 CCD experimental design

Serial Number	Optimization variables					
	Lm1 (mm)	Hm1 (mm)	g (mm)	Wt (mm)	θ (°)	B_{t0} (mm)
1	4.5	21.5	1	5	98.5	1.6
2	3.5	20	1	4	98.5	1.6
3	5.5	20	1	4	98.5	1.6
⋮	⋮	⋮	⋮	⋮	⋮	⋮
81	5.5	21.5	1.2	5	98.5	1.8

Cycle efficiency was evaluated from the weighted representative operating points extracted from the CLTC-P cycle:

$$\eta_{cycle} = \sum_{i=1}^{6} \frac{\lambda_i P_{out,i}}{\lambda_i P_{out,i} + P_{Cu,i} + P_{Fe,i} + P_{PM,i} + P_{mech,i}} \tag{6}$$

Where λ_i is the weight ratio of the i-th representative operating point; $P_{out,\,i}$ represents the mechanical output power at the i-th operating point, which is calculated as the product of the torque and angular speed; and $P_{Cu,\,i}$, $P_{Fe,\,i}$, and $P_{eddy,\,i}$ refer to the copper loss, iron loss, and permanent magnet eddy current loss at each operating point, respectively. In addition, $P_{mech,\,i}$ denotes the mechanical loss.

The loss components were evaluated based on finite element simulations under appropriate control strategies, including maximum torque per ampere and flux-weakening control.

The multi-objective optimization function was determined as follows:

$$F = [\max(T_{avg}), \min(T_{ripple}), \min(AMC), \max(\eta)] \tag{7}$$

4.3 DCO-NSGA-II multi-objective optimization

A response surface model (RSM) based on a central composite design (CCD) was established using finite-element analysis data to approximate the nonlinear relationships between the selected optimization variables and multiple objectives. This surrogate-assisted optimization framework significantly reduces computational cost compared with direct FEA-based optimization, while maintaining sufficient accuracy within the feasible design space. The number of CCD experiments is presented in Table 3.

To illustrate the complex nonlinear relationships between the design variables and the multi-objective performance, the response surfaces with respect to L_{m1} and H_{m1} were studied. The results are shown in Fig. 9, indicating that these variables significantly affect motor torque and material cost.

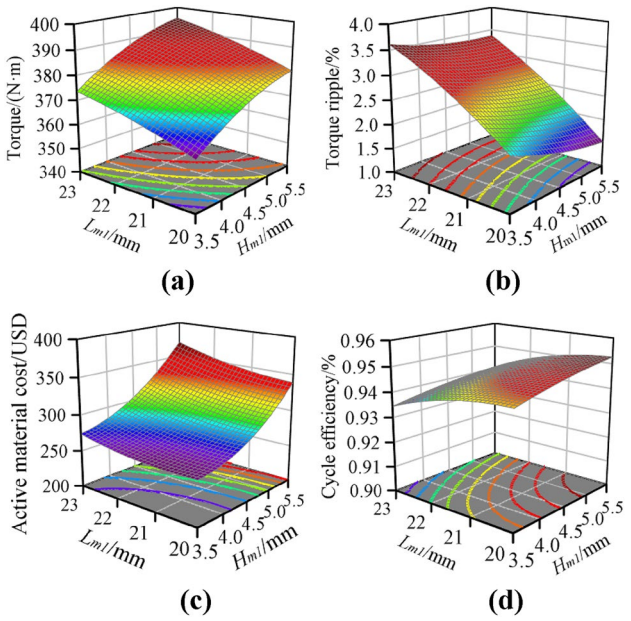


Fig. 9 Objective response surfaces for L_{m1} and H_{m1}: **a** torque; **b** torque ripple; **c** active material cost; **d** cycle efficiency

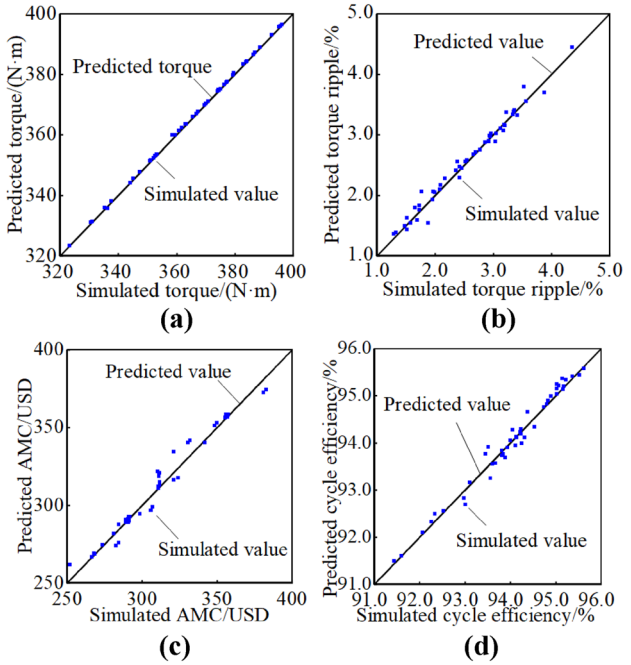


Fig. 10 RSM accuracy validation: **a** torque; **b** torque ripple; **c** active material cost; **d** cycle efficiency

These results show that increasing L_{m1} and H_{m1} generally enhances torque but tends to increase cost and influence torque ripple and efficiency nonlinearly, indicating trade-offs among the objectives.

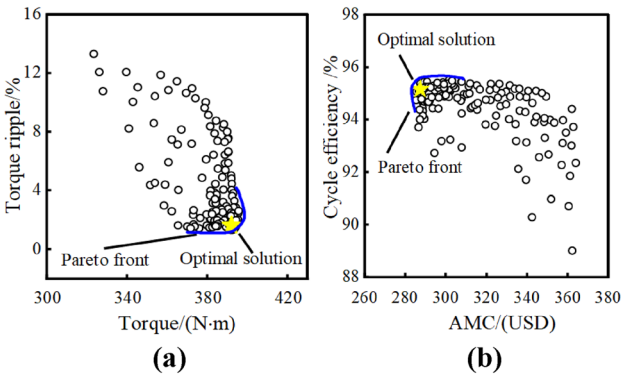


Fig. 11 Pareto optimal set: **a** torque-torque ripple; **b** AMC-cycle efficiency

Table 4 Initial and optimized motor parameter values

Parameter	Initial Value	Optimized Value
Air-gap length g/mm	1	0.8
PM length L_{m1}/mm	22	20.5
PM thickness H_{m1}/mm	5	5.3
PM angle θ_1/(°)	99	98.5
Stator tooth width W_t/mm	15.2	15.6
Slot opening width B_{s0}/mm	1.8	1.5

To evaluate the accuracy of the surrogate model, Fig. 10 compares the predictions from the response surfaces with the high-fidelity electromagnetic simulation benchmarks from the validation set, demonstrating good agreement both quantitatively and qualitatively.

The DCO-NSGA-II improves the classic NSGA-II by incorporating dimension reduction strategies: K-means clustering and Sobol analysis. It identifies the Pareto optimal solution set that balances the four conflicting objectives through non-dominated sorting and crowding distance calculation. The resulting Pareto solution set is shown in Fig. 11.

To select a comprehensive optimal solution from the non-dominated Pareto set, an evaluation function is defined as follows:

$$\min F(P_i) = \varepsilon_1 \frac{T_{avg_initial}}{T_{avg_i}} + \varepsilon_2 \frac{T_{ripple_i}}{T_{ripple_initial}} + \varepsilon_3 \frac{AMC_i}{AMC_{initial}} + \varepsilon_4 \frac{\eta_{initial}}{\mu_i} \tag{8}$$

The optimal solution selection was selected based on practical engineering criteria that prioritize efficiency and while balancing cost and performance requirements for electric vehicle drive motors. The cycle efficiency directly affects the driving range, and the manufacturing cost determines

industrialization feasibility. The torque and torque ripple only need to meet the basic requirements of dynamic performance and driving comfort. Therefore, ε_1, ε_2, ε_3, and ε_4 are set to 0.15, 0.15, 0.35, and 0.35, respectively.

A comparison of motor parameters before and after optimization is presented in Table 4.

5 Simulation results and analysis

Finite element analysis was performed on the amorphous alloy PMSM before and after optimization. To verify the effectiveness of the proposed optimization method, finite element analysis of the optimized PMSM obtained via RSM was conducted for comparison [13].

The output torque performance is shown in Fig. 12. The initial motor achieves a peak torque of 377.5 N·m with a torque ripple of 2.8%. After RSM-based optimization, the peak torque increased to 388.2 N·m, while the torque ripple decreased to 1.7%. In comparison, the motor optimized by the proposed DCO-NSGA-II method achieves the highest peak torque of 390.5 N·m, while further reducing torque ripple to 1.5%.

The breakdown of the active material costs for the three motor designs discussed in this paper is summarized in Table 5. The total active material cost of the initial design is 294.72 USD, while the RSM-based optimization design costs 291.86 USD. For the proposed DCO-NSGA-II design, the total active material cost is further reduced to 291.13 USD, representing a net cost reduction of 3.59 USD relative to the initial design. This is mainly due to the optimized usage of permanent magnet and silicon steel.

Efficiency maps of the three designs are compared in Fig. 13. The initial motor exhibits a driving-cycle efficiency of 94.0%, with the high-efficiency region (efficiency ≥ 90%) accounting for 80.33% of the operating area. The RSM-optimized design shows a moderate expansion of the high-efficiency region. The proposed DCO-NSGA-II design achieves a driving-cycle efficiency of 95.5%, representing an improvement of 1.5% points over the initial design. In addition, the proportion of the high-efficiency operating region increases to 85.39%, indicating a more favorable efficiency distribution over a wide range of operating conditions.

The comparative results demonstrate that conventional RSM-based optimization can improve specific performance indicators to a certain extent. However, its effectiveness is limited when multiple conflicting objectives and full driving-cycle considerations are involved. In contrast, the proposed DCO-NSGA-II optimization method explicitly incorporates driving-cycle weighting and maintains solution diversity via Pareto dominance and crowding-distance mechanisms. This enables coordinated improvements in torque performance, efficiency distribution, and material cost.

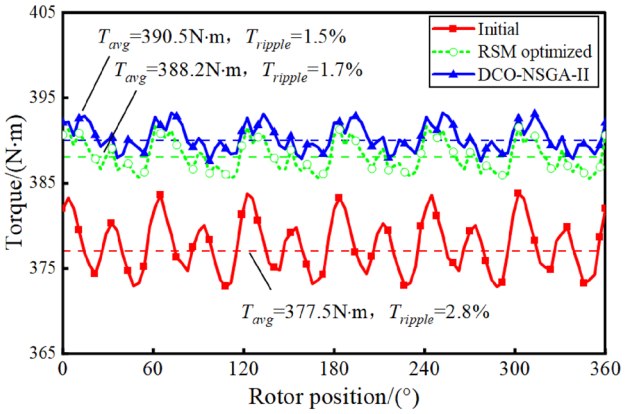


Fig. 12 Comparison of motor output performance

Table 5 Comparison of active material costs

	Amorphous Alloy (USD)	Permanent Magnet (USD)	Copper Wire (USD)	Silicon Steel (USD)	Total (USD)
Initial	54.34	140.13	63.06	37.18	294.72
RSM optimized	56.63	137.85	61.78	35.60	291.86
DCO-NSGA-II	56.48	136.84	61.78	36.03	291.13

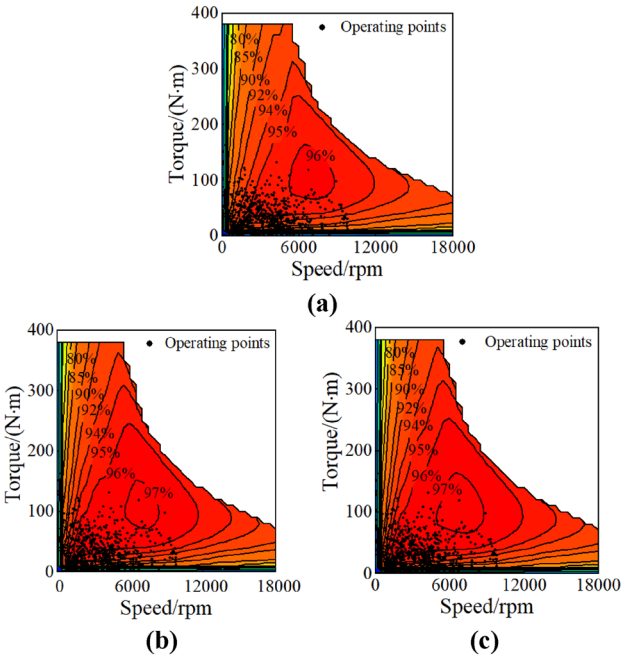


Fig. 13 Comparison of motor efficiency maps: **a** initial efficiency map; **b** RSM optimized efficiency map; **c** DCO-NSGA-II efficiency map

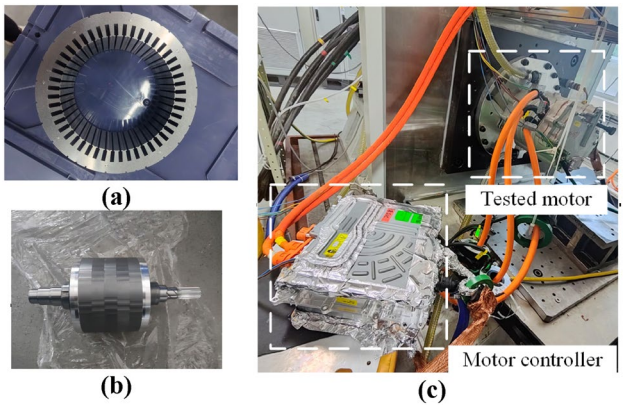


Fig. 14 Prototype and experimental platform: **a** stator; **b** rotor; **c** experimental test bench

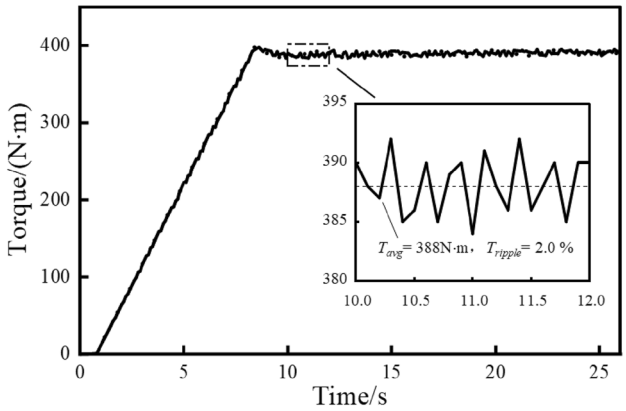


Fig. 15 Verification of motor torque performance

6 Experimental results

To validate the accuracy of the simulation results and the feasibility of the proposed optimization, a prototype amorphous alloy PMSM was manufactured according to the optimized parameters. Figure 14 shows the fabricated amorphous alloy stator core, rotor, and experimental test setup.

During the torque performance evaluation, the prototype was powered by a controller with a rated DC bus voltage of 630 V, and the dynamometer maintained a constant speed of 3000 rpm. The motor current was gradually increased to elevate the output torque until the peak current was reached. The torque was measured using a high-precision sensor with a sampling interval of 0.1 s. The torque performance of the prototype motor was evaluated under peak operating conditions. Figure 15 presents the measured torque response during a step loading test. The experimental results show that the prototype achieves a peak torque of 388 N·m with a torque ripple of approximately 2.0%. Compared with the simulation results, the measured peak torque exhibits a slight deviation, which can be attributed to manufacturing tolerances, material property variations, and measurement

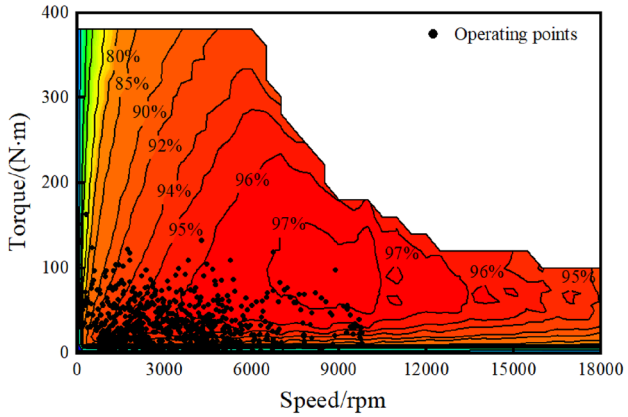


Fig. 16 Experimental verification of motor efficiency map

uncertainties. Nevertheless, the overall agreement between the experimental and simulated torque characteristics confirms the reliability of the electromagnetic model and the effectiveness of the optimized design.

The reported efficiency corresponds exclusively to the motor-only efficiency. The electrical input power was measured using a high-precision power analyzer, while the mechanical output was captured via torque and speed sensors. The efficiency map was constructed by sampling test points at intervals of 200 rpm below 1000 rpm and at intervals of 500 rpm above 1000 rpm, with load increments of 20 N·m up to the peak torque. To prevent thermal accumulation from affecting measurement accuracy, a 5-minute no-load cooling interval was strictly enforced between high-load tests.

The efficiency map of the prototype amorphous alloy PMSM was obtained through steady-state efficiency measurements across a wide range of speeds and loads, as illustrated in Fig. 16. Based on the measured efficiency map, the driving-cycle efficiency under the CLTC-P operating conditions was calculated to be 95.3%. This experimental value is in close agreement with the simulated driving-cycle efficiency of 95.5%, with a slight deviation of less than 0.3%. The consistency between the experimental and simulation results indicates that the loss models and efficiency evaluation method employed in this study are appropriate for performance assessment under full operating conditions.

7 Conclusion

A new amorphous alloy PMSM and a corresponding multi-objective optimization method to meet the requirements of electric vehicles for superior performance under full operating conditions and cost effectiveness were proposed in this paper. By integrating the K-means clustering of the CLTC-P cycle with Sobol sensitivity analysis into DCO-NSGA-II

optimization, the proposed method effectively coordinates conflicting objectives. The optimized amorphous alloy PMSM design achieves a 3.44% increase in torque, a 1.3% reduction in torque ripple, a 1.5% boost in cycle efficiency, and reduced material costs. Both simulation and experimental results confirm the effectiveness of the proposed amorphous alloy PMSM as well as the DCO-NSGA-II optimization method for electric vehicle applications. This research may enhance the competitiveness of amorphous alloy PMSMs in the electric vehicle market.

Declarations

Conflict of interest The authors declare that this paper has no problems in terms of conflict of interest, ethicals, informed consent, data availability, or funding.

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